Multiple Linear Regression

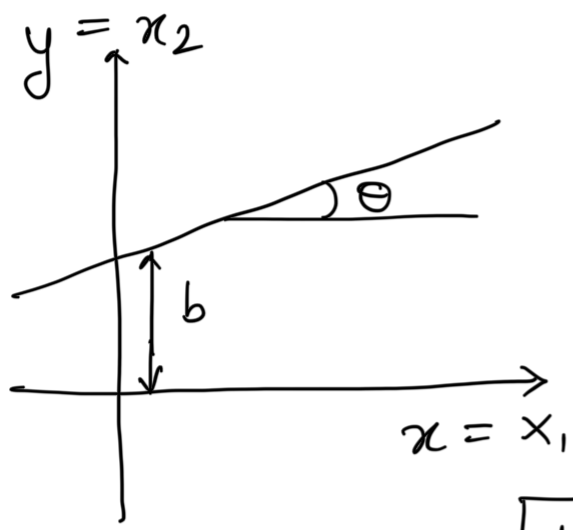
$$y = mx_1 + mx_2 + \dots mx_n + b$$

OR

$$\boxed{y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n}$$

$$\therefore \boxed{y = \beta_0 + \sum \beta_i x_i}$$

Equation of Hyperplane



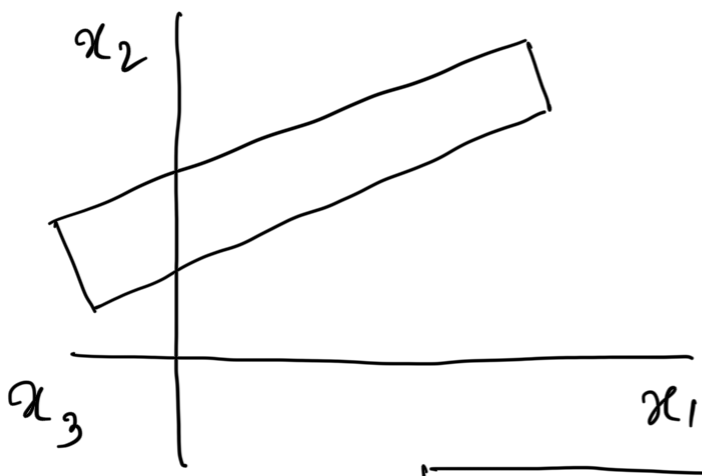
$$y = mx + c$$

which is similar to

$$ax + by + c = 0$$

$$\Rightarrow \boxed{w_1 x_1 + w_2 x_2 + w_0 = 0}$$

This is the equation of line



$$\Rightarrow \boxed{w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0}$$

This is the equation of plane.

$$\boxed{w_1x_1 + w_2x_2 + w_3x_3 + \dots + w_nx_n + w_0 = 0}$$

This is the equation of hyperplane for n-dimensional co-ordinal space.

Hyperplane is denoted as π in higher dimensions

Equation of line in vector format

$$w = [w_1, w_2, w_3 \cdots w_n]$$

$$x = [x_1, x_2, x_3 \cdots x_n]$$

$$w^T = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \quad \therefore \quad w^T x = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} [x_1, x_2 \cdots x_n]$$

$$a \cdot b = a^T b$$

$$\therefore \boxed{w^T x + w_0 = 0}$$

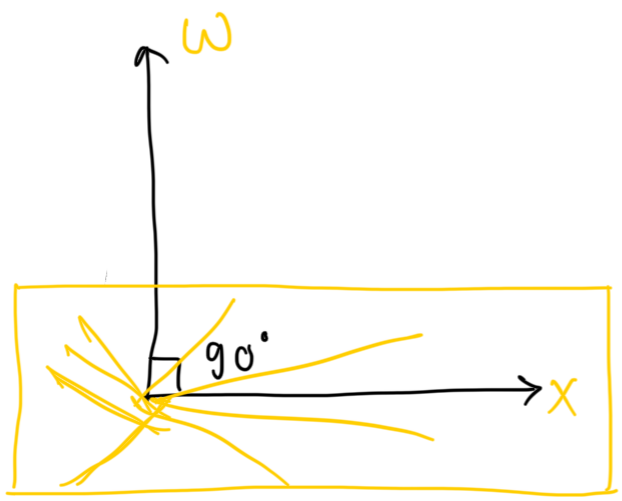
If hyperplane passes through the origin then w_0 becomes zero

$$\therefore \boxed{w^T x = 0}$$

For n dimensions

$$w_1x_1 + w_2x_2 + \dots w_nx_n = 0$$

$$w^Tx = 0$$

$$w^Tx = w \cdot x = \|w\| \, \|x\| \cos\theta = 0$$

here $\theta = 90^\circ$